

Interesting Concepts :

1. Images are just numbers Every image is simply a grid of numbers representing pixel values. Computers don't "see" pictures they process matrices. This was a surprisingly simple but mind-shifting idea.
2. Low, Mid, and High Level Vision Computer vision tasks are organized in levels from basic pixel operations (low), to understanding motion and depth (mid), to recognizing objects and scenes (high). Real systems build upward through all three.
3. How Convolutional Filters Work Small filters slide across an image to detect patterns like edges and textures. The cool part is that neural networks *learn* these filters automatically during training rather than having them manually designed.
4. Optical Flow By tracking how pixels shift between video frames, machines can perceive motion. A simple but powerful idea the brightness of a moving point stays roughly constant, and that assumption alone enables motion tracking.
5. YOLO You Only Look Once Developed by Redmon himself, YOLO detects objects in a single pass through the network instead of multiple stages. It made real-time object detection practical, which was a big breakthrough.

Healthcare has several types of reports lab reports, discharge summaries, progress notes, operative reports, and medical imaging reports. A medical imaging report (also called a radiology report) is written by a radiologist after analyzing scans like X-rays, MRIs, CT scans, or ultrasounds.

Structure of a Medical Imaging Report:

- Patient Info & Clinical Indication Basic patient details and *why* the scan was ordered
- Type of Exam & Technique Which modality was used and how (e.g., contrast CT vs. plain MRI)
- Comparison Whether prior scans are available to track changes over time
- Findings A systematic description of what was observed in each anatomical region
- Impression A concise summary and likely diagnosis; this is what the referring doctor reads first